

Deep Learning and Econometric Models for Volatility Forecasting

Advanced Financial Time-Series Analysis

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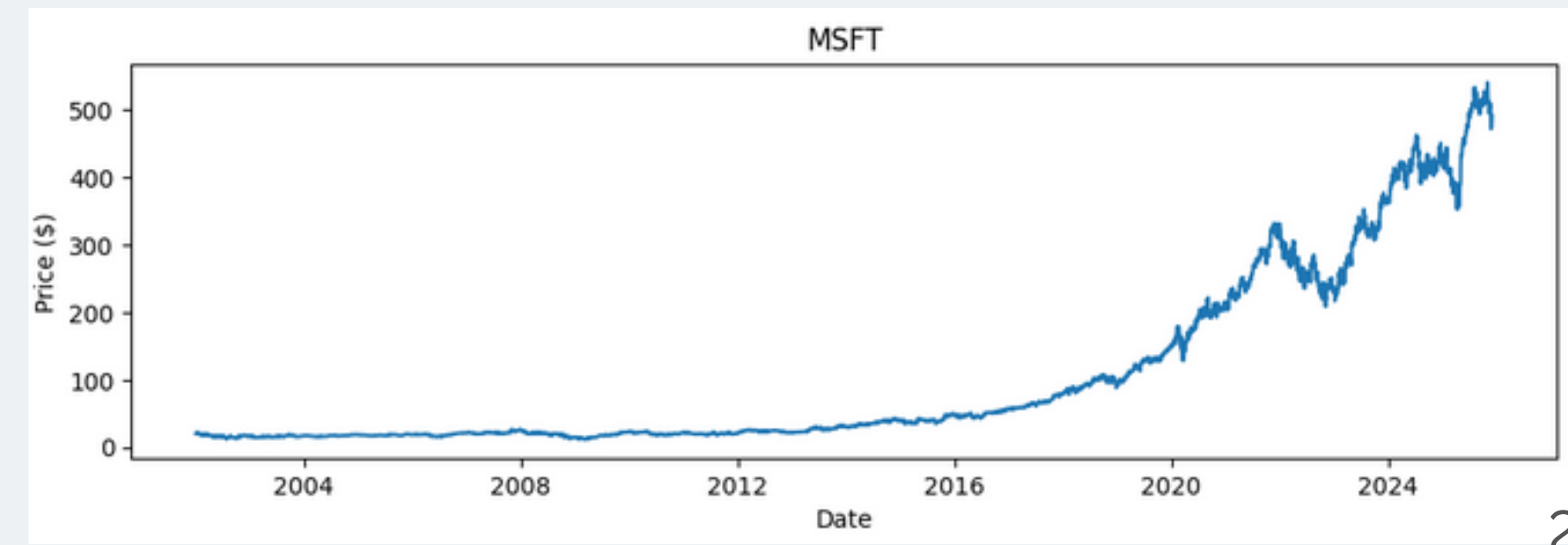
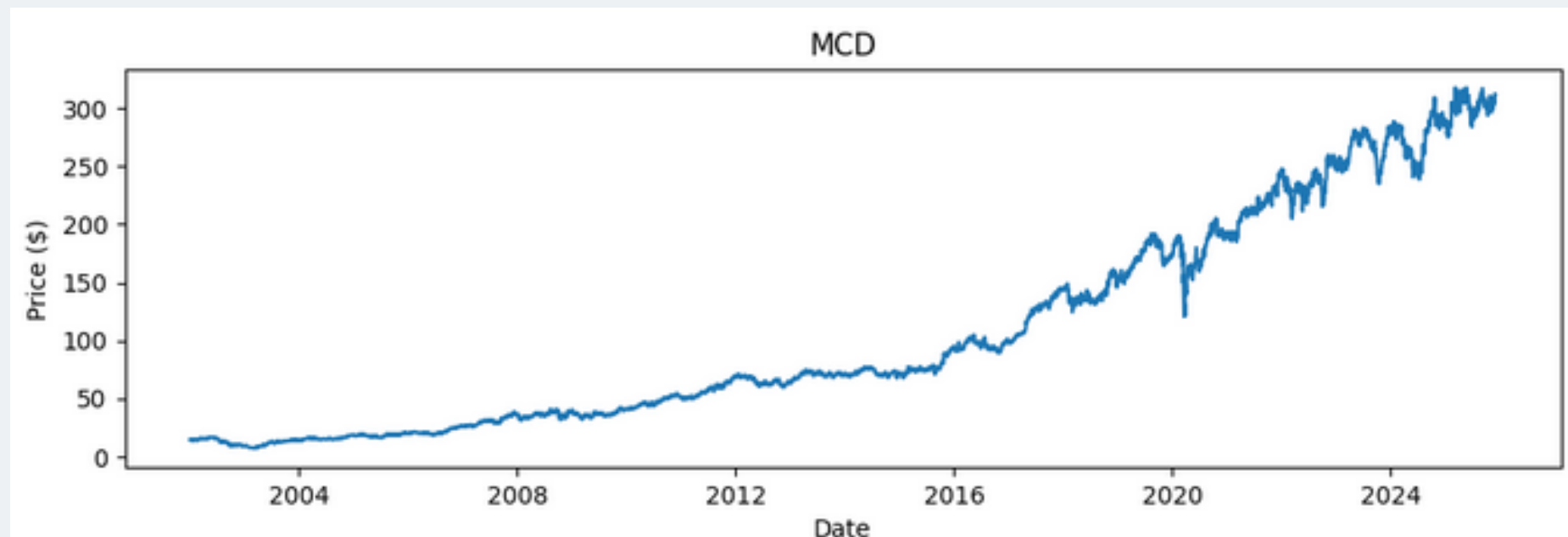
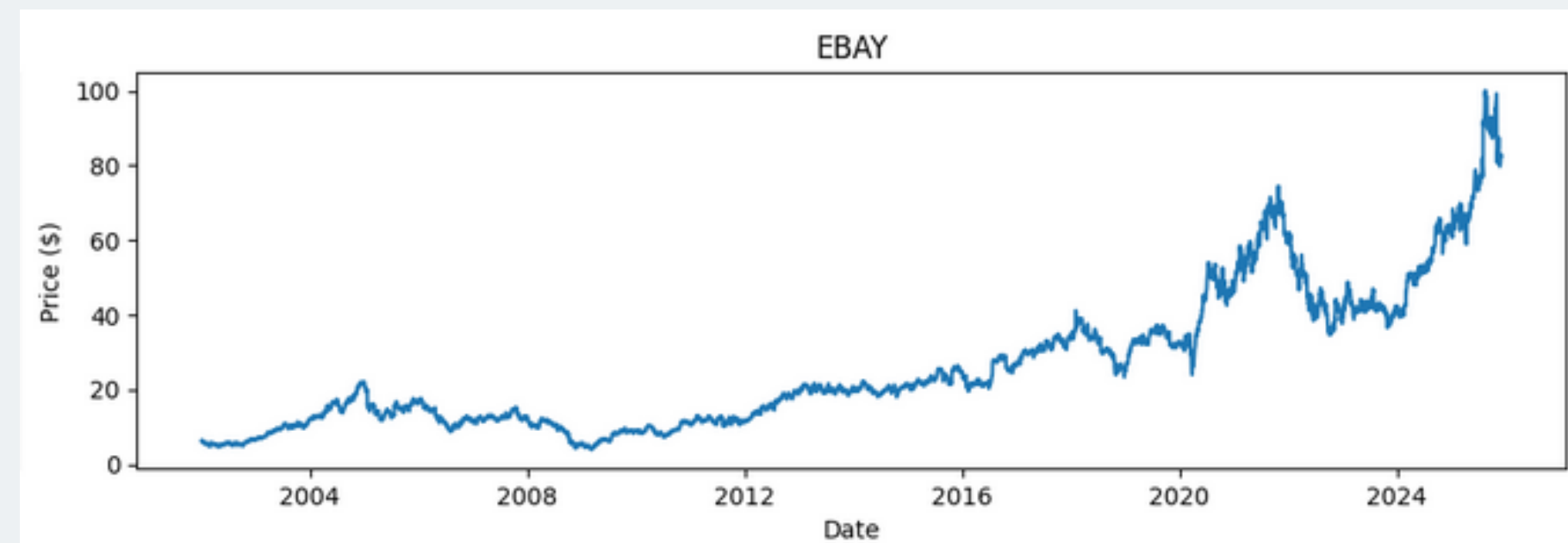
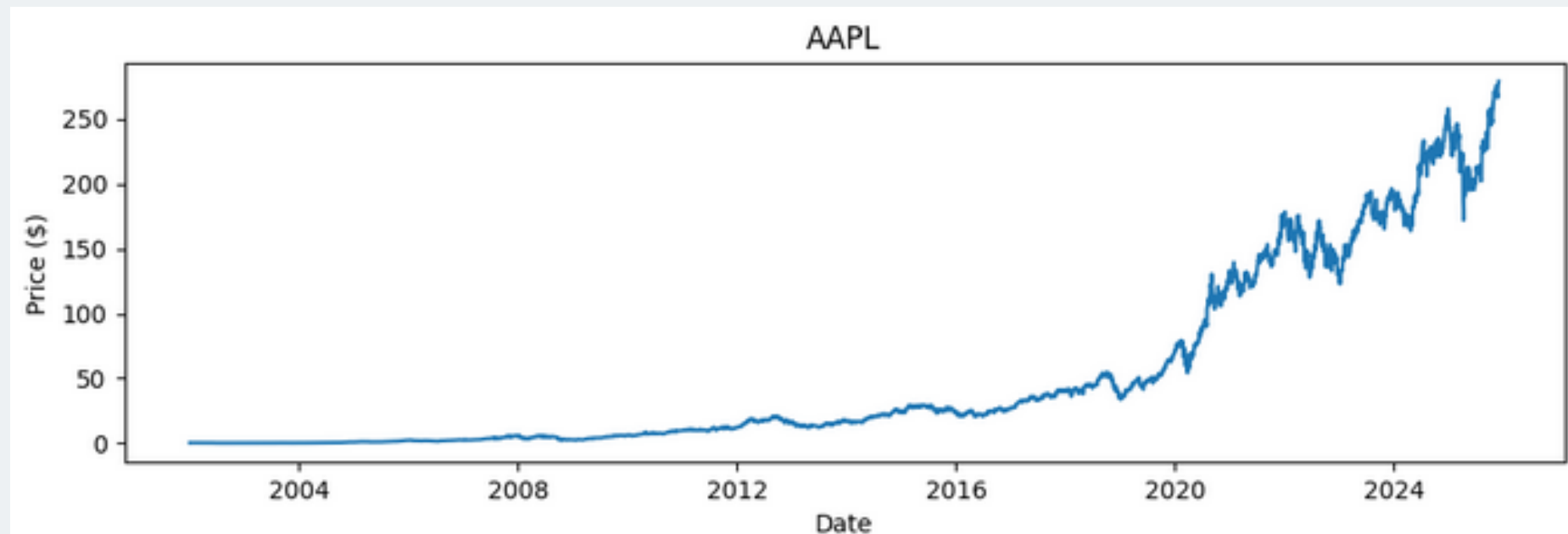
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CIS ID: VA5698892 Artur Vasilian



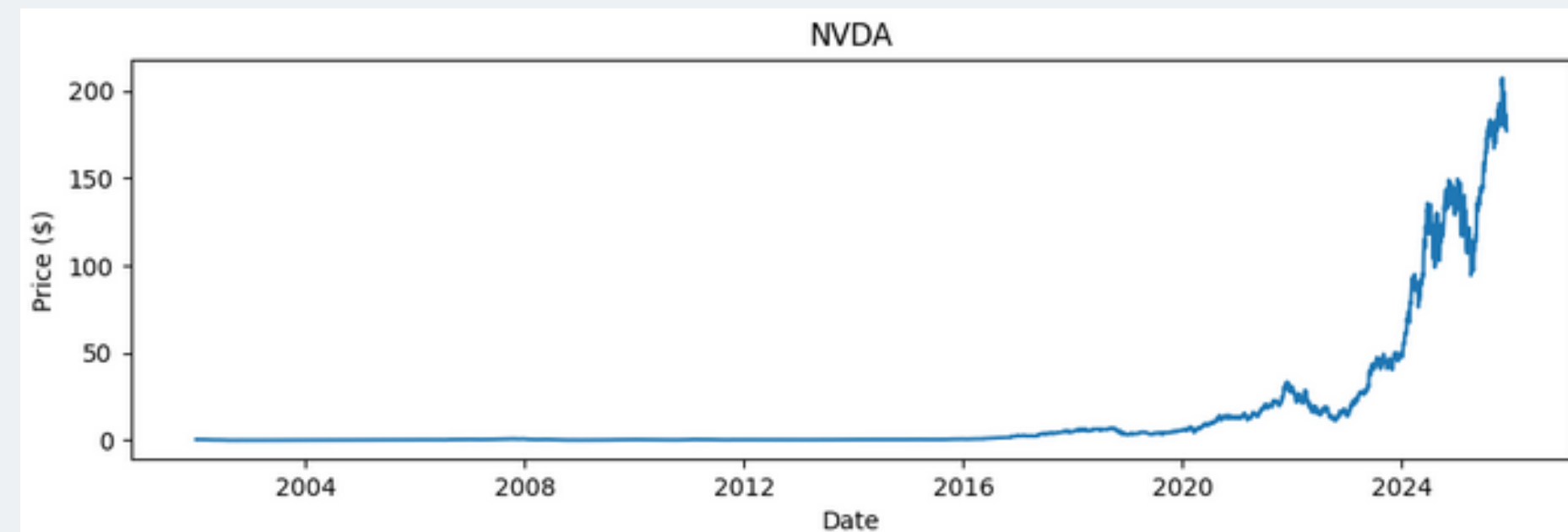
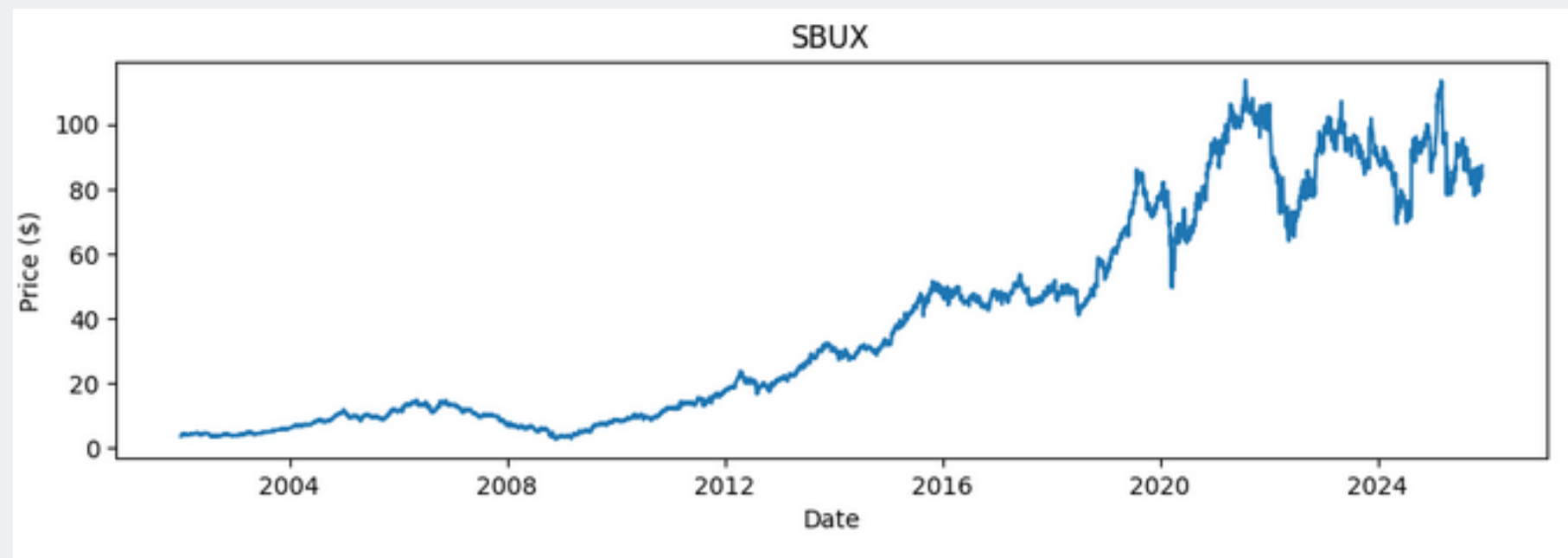
Problem statement

Financial markets are unpredictable, and different models capture different volatility patterns. We compare GARCH, deep learning, and hybrid methods to find the most accurate volatility forecasting approach.



Why the problem matters

- Traditional econometric models struggle with nonlinear and structural changes in markets.
- Deep learning models offer new capabilities, but it is unclear when they outperform classical models.
- Comparing GARCH with LSTM/Transformer models helps fill an important gap in volatility forecasting research.
- Better volatility forecasts reduce portfolio risk and improve trading, hedging, and risk-management decisions.
- Financial markets are increasingly unstable due to economic shocks, geopolitical events, and rapid information flows.



What is being solved ?

Evaluate whether hybrid models outperform single-method approaches.

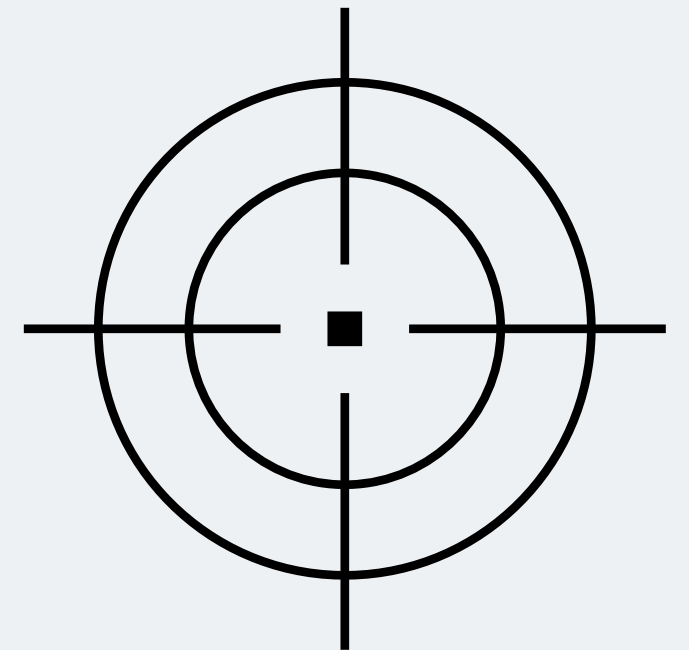
Test if combining econometric structure (GARCH) with deep-learning flexibility (LSTM/Transformer) improves forecasting accuracy.

Identify the most accurate volatility forecasting model.

Compare GARCH, LSTM, Transformer, and Hybrid models to determine which achieves the lowest prediction error.

Assess whether deep learning captures market patterns missed by classical models.

Determine if modern neural models provide meaningful gains over traditional econometric volatility models.



Dataset overview



- **Source:** Yahoo Finance
- **Domain:** Financial time-series (daily stock prices)
- **Data Type:** Historical data (Close)
- **Tickers Used:**
 - AAPL
 - EBAY
 - MCD
 - MSFT
 - NVDA
 - SBUX
- **Period Covered:** 01/01/2002–30/11/2025

Ticker	AAPL	EBAY	MCD	MSFT	NVDA	SBUX
Date						
1/2/2002	0.349395	6.196217	14.364336	20.475895	0.51422	3.642587
1/3/2002	0.353594	6.36758	14.527014	21.144781	0.547915	3.812486
1/4/2002	0.355244	6.4144	14.635463	21.043991	0.531947	4.128017
1/7/2002	0.343398	6.255211	14.74934	20.940153	0.517658	4.077605
1/8/2002	0.339049	6.324505	14.836093	21.190594	0.523771	4.236305

Ticker	AAPL	EBAY	MCD	MSFT	NVDA	SBUX
count	6017	6017	6017	6017	6017	6017
mean	51.543892	25.941941	106.210667	104.32972	14.594098	39.176666
std	69.637333	18.689384	90.507583	132.256638	36.102397	33.009765
min	0.196741	3.846767	6.806715	11.138551	0.056312	2.67733
25%	3.588428	11.675143	33.470512	19.069792	0.295715	9.6983
50%	18.123589	20.046642	70.905342	30.499249	0.50465	29.163971
75%	62.743656	35.616306	172.205963	141.994934	6.308578	69.035515
max	278.850006	100.101936	317.67453	541.057373	207.039993	113.5653

Data description

~6,000 + daily observations per stock
(2002–2025)

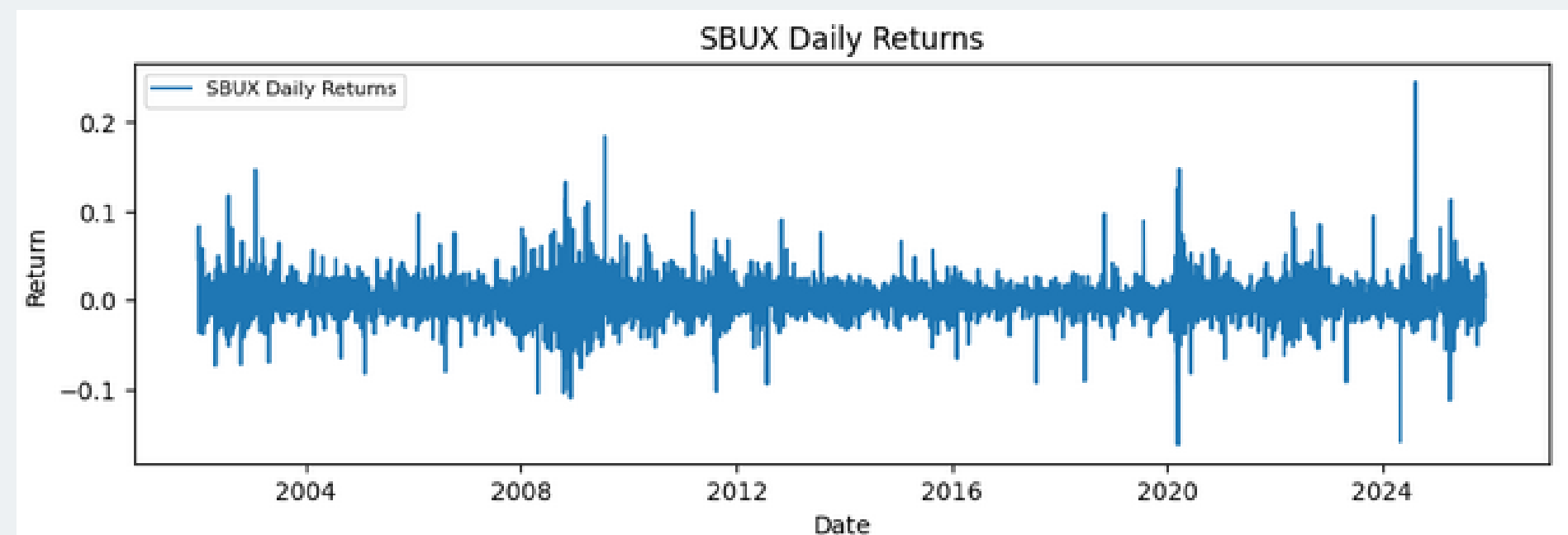
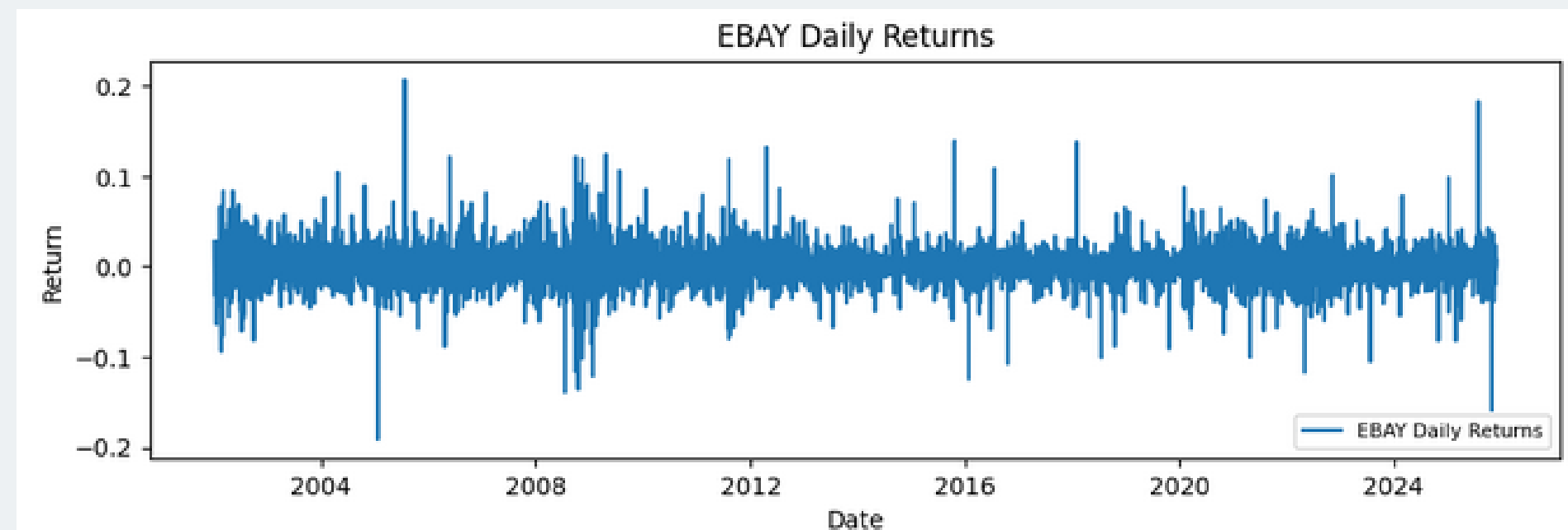
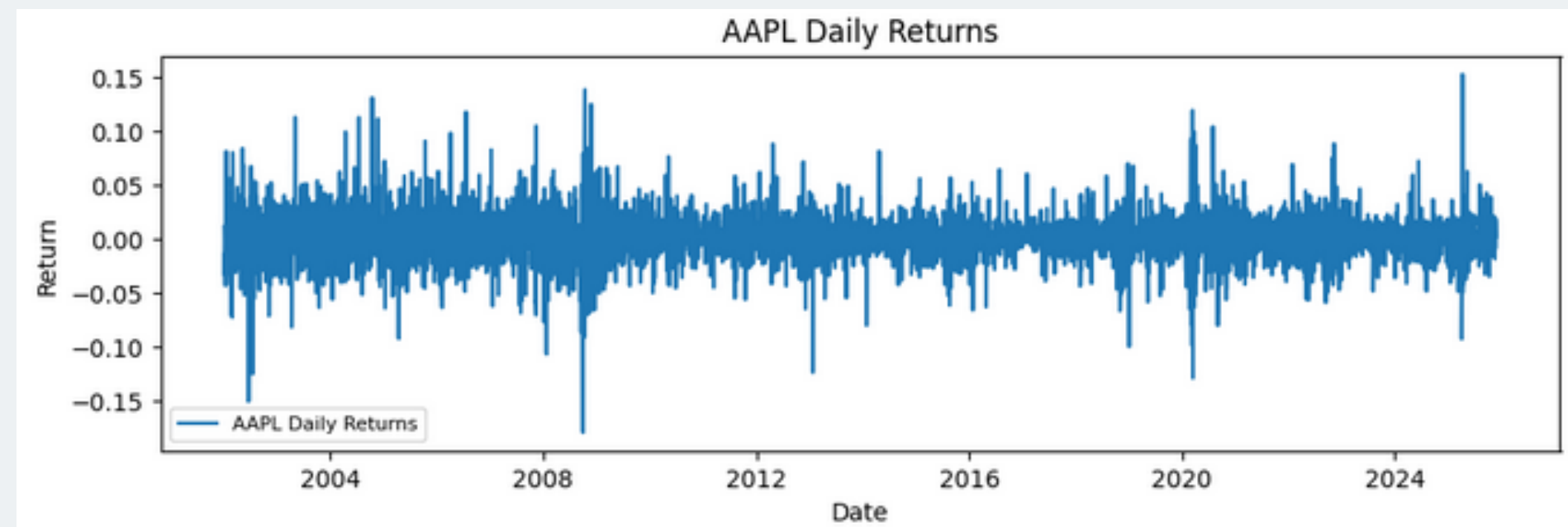
Realized volatility

Daily log returns

$$z_t = \log \left(\frac{p_t}{p_{t-1}} \right) .$$

$p(t)$ _ the price of an asset at time t .

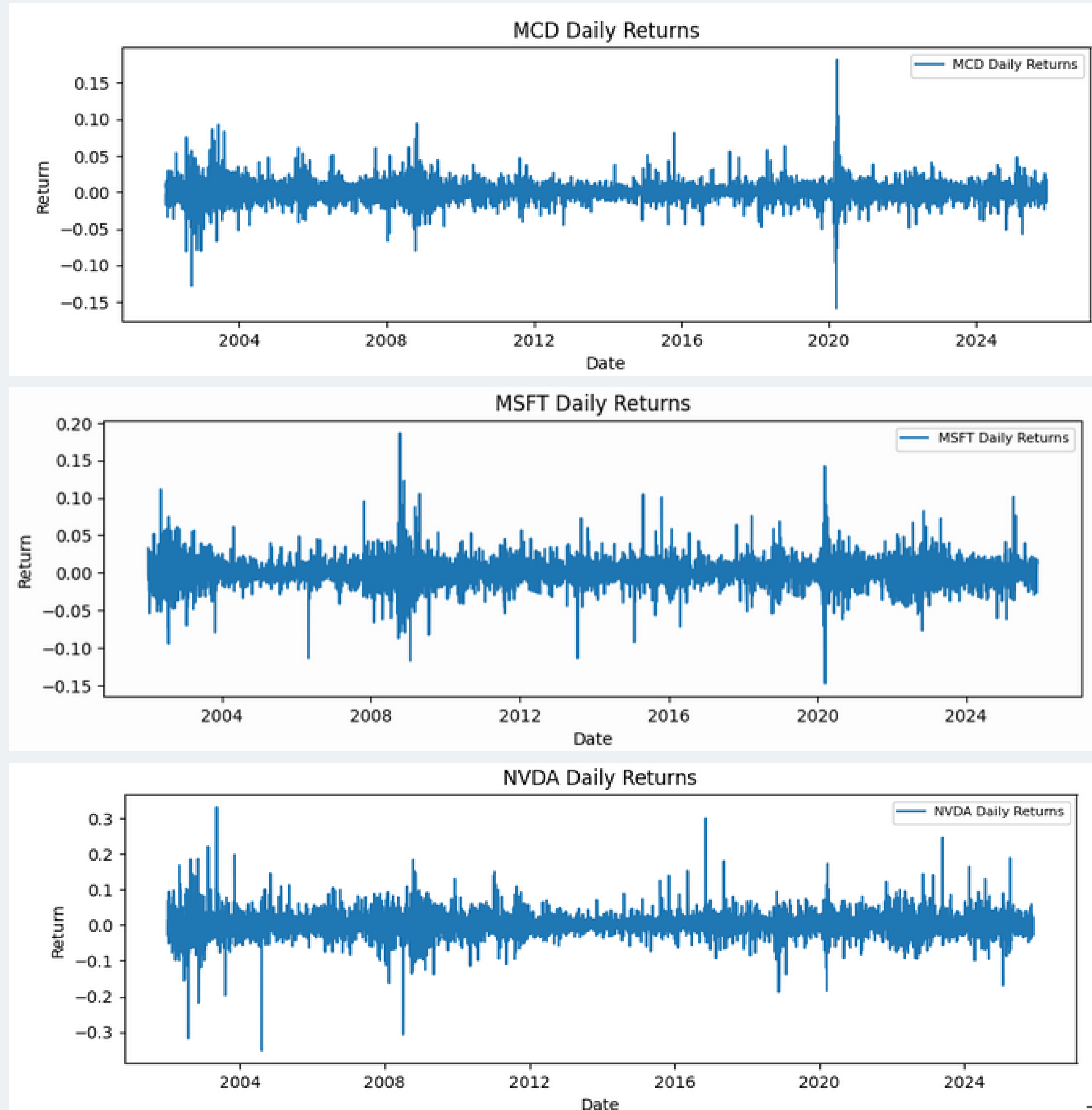
- Squared returns (for volatility clustering)
- Rolling Std (30-day)
- GARCH conditional variance
- Scaled return sequences (for LSTM / Transformer models)



Data description

Returns show heteroskedasticity and clustered volatility typical of financial time series.

This motivates the use of ARCH/GARCH and deep learning models for accurate volatility forecasting.



Methodology model families and key formulas

ARCH Models (Econometric).

$$\alpha_0 + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2,$$

GARCH Models (Econometric)

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

Hybrid Model (GARCH + Deep Learning)

Convex Blend (simple hybrid)

$$\hat{v}_{t+1}^{Hybrid} = w \cdot \hat{v}_{t+1}^{DL} + (1 - w) \cdot \hat{v}_{t+1}^{GARCH}, \quad 0 \leq w \leq 1$$

Linear regression hybrid

$$\hat{v}^{Hybrid} = \beta_1 \hat{v}^{LSTM} + \beta_2 \hat{v}^{GARCH}$$

Non-negative convex weights

$$w_i \geq 0, \quad \sum w_i = 1$$

LSTM (Long Short-Term Memory)

Gate Name	Equation
Input Gate	$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$
	$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$
Forget Gate	$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$
Update Gate	$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$
	$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$
Output Gate	$h_t = o_t \odot \tanh(c_t)$

Methodology

ARCH/GARCH – Traditional Econometric Models

- Capture volatility clustering
- Volatility depends on past shocks
- Strong short-term forecasts

LSTM – Deep Learning Model

- Sequence model for time series
- Learns long-term patterns
- Handles non-linear behavior

Transformer – Attention-Based Deep Learning

- Focuses on important time steps
- Handles long sequences well
- Learns complex volatility patterns

Hybrid Models – Combined Approach

- Combine GARCH + deep learning
- Methods: weighted blend, ridge, OLS, AdaBoost
- Aim: improve accuracy using both models

Baseline – Rolling Standard Deviation

- 30-day rolling standard deviation
- Simple historical volatility benchmark

Evaluation metrics

Primary Metric: RMSE (Root Mean Squared Error)

- Measures average deviation between predicted volatility and realized volatility
- Penalizes larger errors more strongly → good for risk applications
- Lower RMSE = more accurate volatility forecast

Why RMSE?

- Standard metric in volatility forecasting literature
- Easy to compare across models
- Sensitive to prediction spikes and extreme errors (important for finance)

Data preprocessing and modeling setup

Returns and Cleaning

- Used daily log returns and converted to percent ($\times 100$) for GARCH
- Aligned all tickers on the same dates
- Ensured no future leakage in time alignment

Volatility Target

- Target: absolute daily return $|r_t|$ as a proxy for realised volatility

Train/Test Split

- For hybrid 70% train / 30% test, for the individual methods 80% train / 20% test
- GARCH: rolling expanding window, refit daily
- LSTM / Transformer: trained once, then sliding-window prediction on test set

Model Inputs

- ARCH/GARCH: daily percent returns only, LSTM: 10-day windows of past volatility
- Hybrid: combined LSTM, best GARCH, and rolling std baseline predictions as features

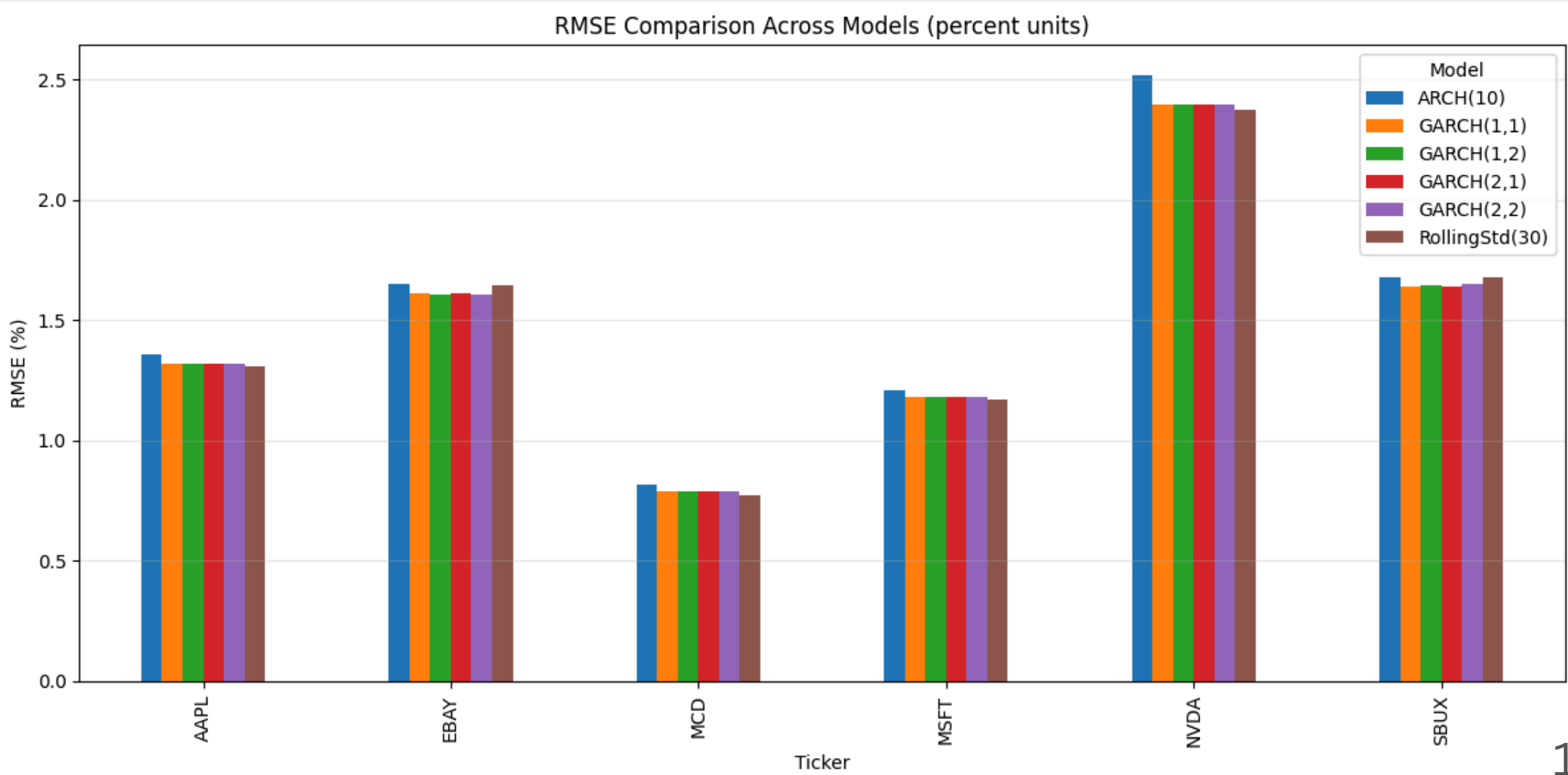
Hardware

- Experiments run in Google Colab / standard GPU environment.

Results: Econometric Models (GARCH)

- GARCH models outperform the 30-day rolling baseline across majority of stocks.
- GARCH(1,1) and GARCH(1,2) provide the best overall RMSE, with only small differences between them.
- ARCH(10) is the weakest, confirming the importance of including volatility persistence.
- GARCH captures broad volatility patterns but fails to predict sharp spikes, especially for high-volatility stocks.

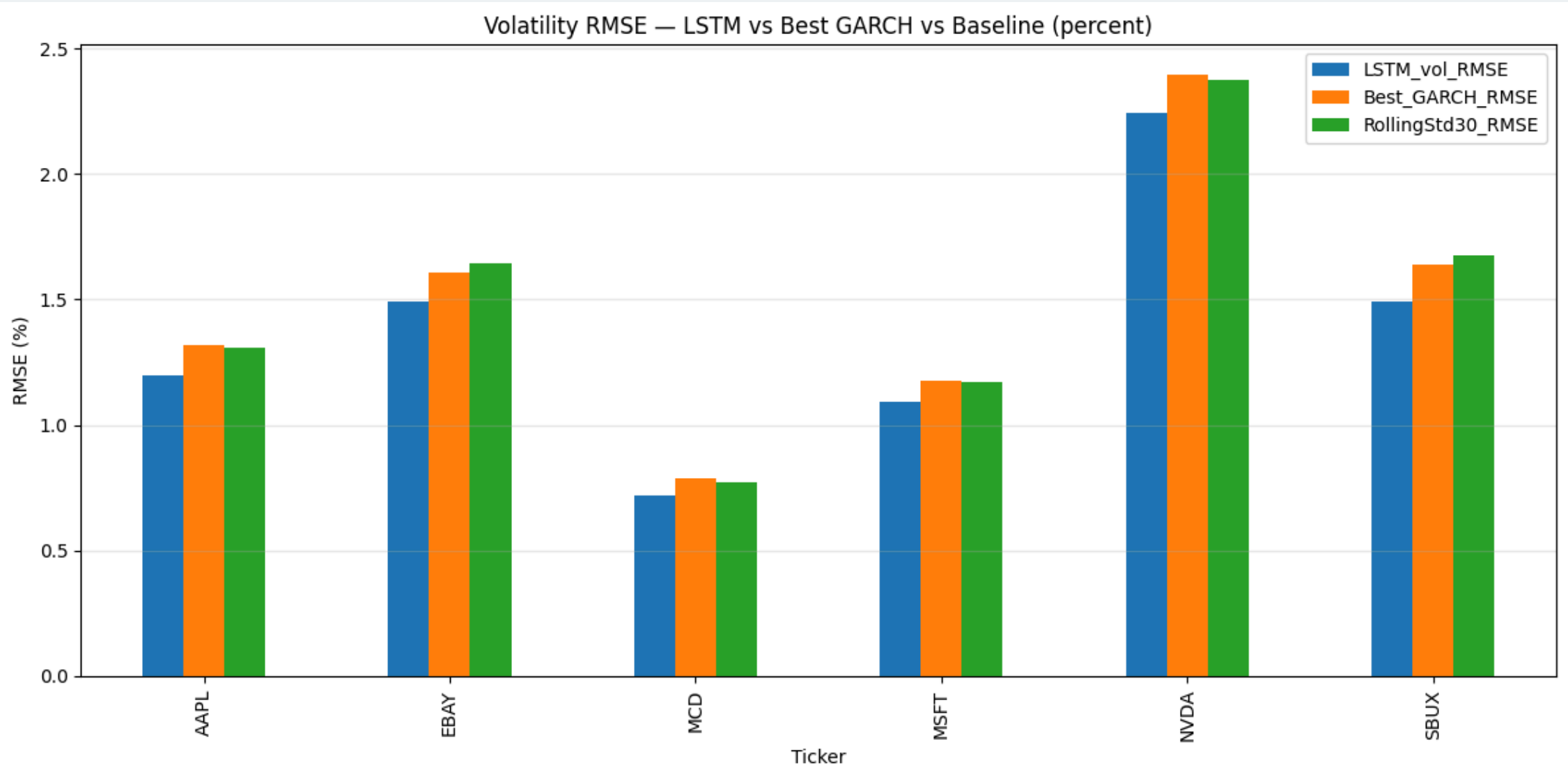
Model/Ticker	ARCH(10)	GARCH(1,1)	GARCH(1,2)	GARCH (2,1)	GARCH (2,2)	Rolling std(30)
AAPL	1.357	1.322	1.319	1.322	1.319	1.307
EBAY	1.653	1.613	1.608	1.613	1.609	1.647
MCD	0.814	0.788	0.788	0.788	0.788	0.773
MSFT	1.208	1.179	1.179	1.179	1.179	1.170
NVDA	2.518	2.395	2.398	2.395	2.398	2.377
SBUX	1.677	1.642	1.643	1.642	1.651	1.678



Results: LSTM Model

- LSTM achieves the lowest RMSE across all six stocks, outperforming the baseline and best GARCH model by 6–11%.
- It produces smooth, stable next-day volatility forecasts, learning nonlinear and regime-shift patterns that GARCH cannot, and—despite missing extreme spikes—remains the most accurate standalone model in the study.

Ticker	LSTM volatility RMSE (percent units)
AAPL	1.200137
EBAY	1.490219
MCD	0.721927
MSFT	1.093799
NVDA	2.242718
SBUX	1.49063

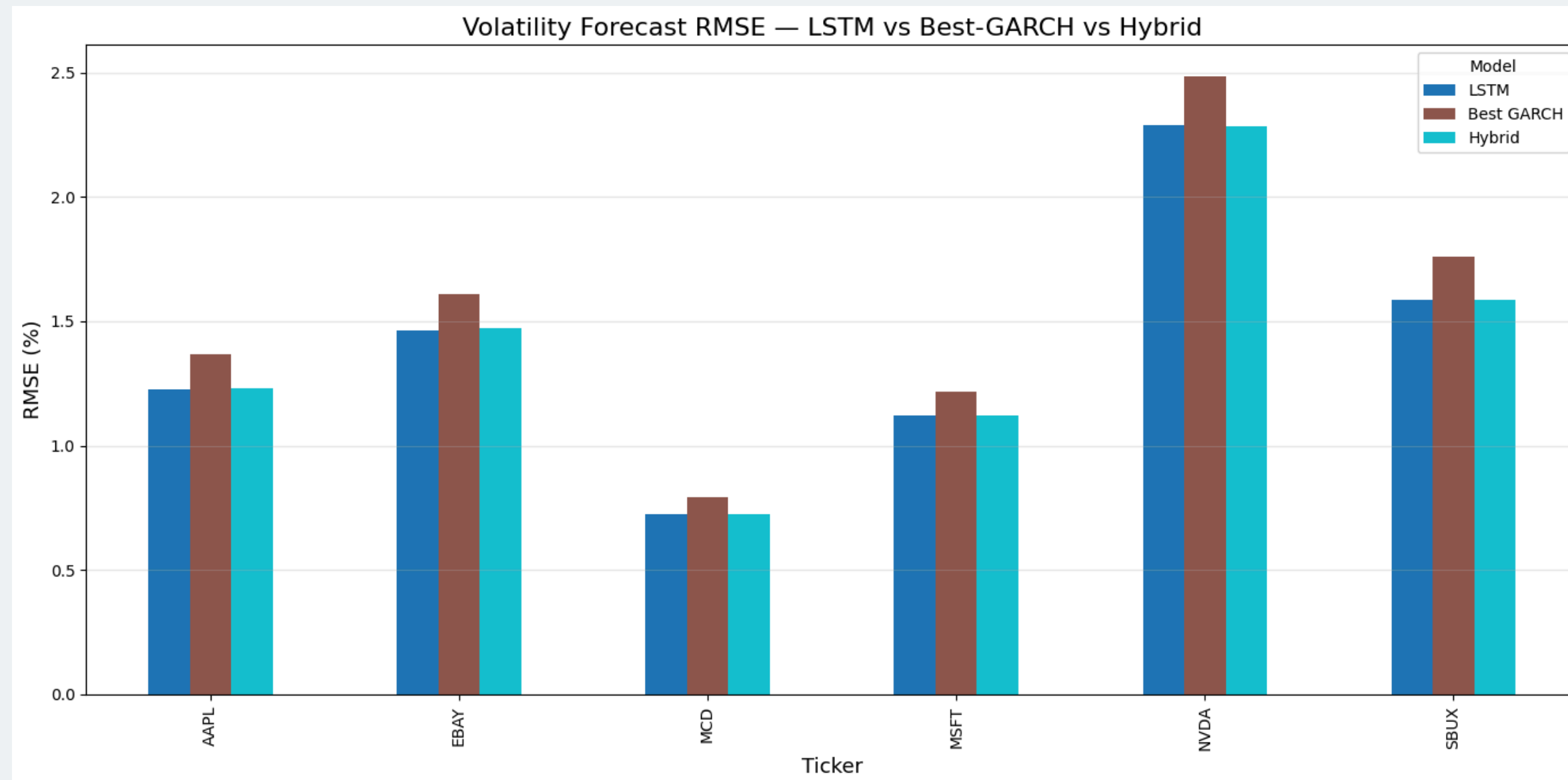


Results: Hybrid Models-Convex blend

Option A — Convex blend (simple & robust) Constraint weights to a convex combo that minimises MSE on a fit slice (first 70% of your train window), then evaluate on the remaining 30%.

Convex Blend (simple hybrid)

$$\hat{v}_{t+1}^{Hybrid} = w \cdot \hat{v}_{t+1}^{DL} + (1 - w) \cdot \hat{v}_{t+1}^{GARCH}, \quad 0 \leq w \leq 1$$



Results: Hybrid Models-Convex blend

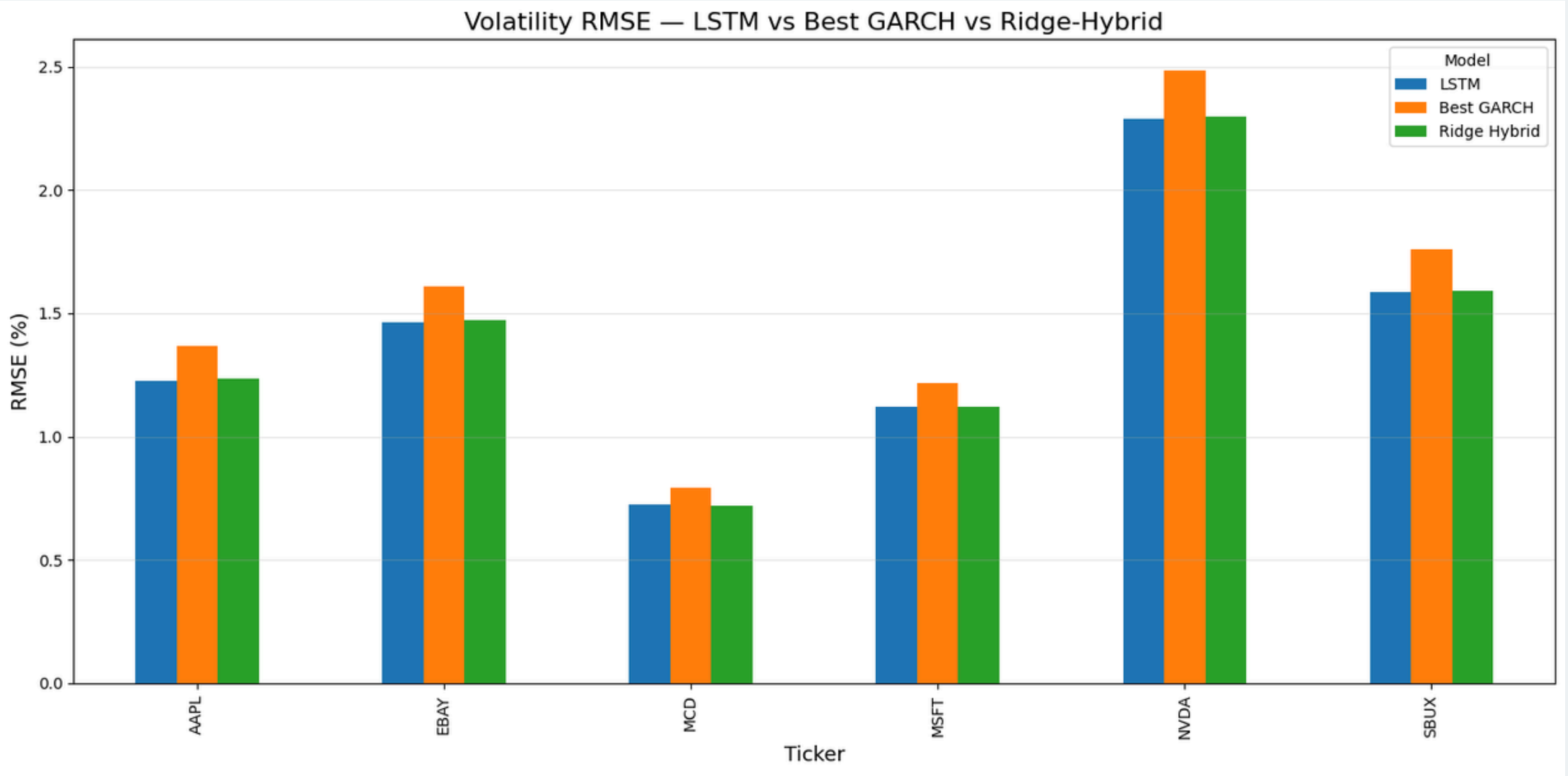
- Hybrid models combine LSTM and GARCH forecasts with optimized weights, typically giving more weight to LSTM (60–95%) across all stocks.
- The hybrid approach consistently improves over GARCH and achieves RMSE very close to the LSTM model.
- For some stocks (e.g., NVDA, AAPL), the hybrid slightly outperforms LSTM, showing benefit from adding GARCH’s volatility-clustering structure.
- Overall, the convex blend is stable, simple, and robust, providing the best balance between deep-learning flexibility and econometric interpretability.

AAPL: Best GARCH=GARCH(1,2), w_lstm=0.67, w_garch=0.33 eval RMSE — LSTM 1.2279, GARCH 1.3672, HYBRID 1.2312
MSFT: Best GARCH=GARCH(2,2), w_lstm=0.78, w_garch=0.22 eval RMSE — LSTM 1.1206, GARCH 1.2176, HYBRID 1.1228
NVDA: Best GARCH=GARCH(1,1), w_lstm=0.56, w_garch=0.44 eval RMSE — LSTM 2.2880, GARCH 2.4861, HYBRID 2.2827
EBAY: Best GARCH=GARCH(1,2), w_lstm=0.64, w_garch=0.36 eval RMSE — LSTM 1.4644, GARCH 1.6081, HYBRID 1.4716
MCD: Best GARCH=GARCH(1,1), w_lstm=0.97, w_garch=0.03 eval RMSE — LSTM 0.7252, GARCH 0.7924, HYBRID 0.7253
SBUX: Best GARCH=GARCH(2,1), w_lstm=0.74, w_garch=0.26 eval RMSE — LSTM 1.5842, GARCH 1.7574, HYBRID 1.5877

Results: Hybrid Models-Ridge regression

- The ridge-based hybrid consistently improves over the best GARCH model on all stocks ($\approx +7\%$ to $+9.5\%$ RMSE improvement).
- Compared to LSTM, the hybrid delivers very similar accuracy, usually within $\pm 0.5\%$, confirming stable performance across assets.
- Ridge regression tends to lean more heavily on GARCH weights, but still benefits from adding a small LSTM component.
- Overall, the ridge hybrid is robust and reliable, offering a strong middle ground between GARCH's stability and LSTM's flexibility.

Improvements (%):		
	Hybrid_vs_LSTM_impr_%	Hybrid_vs_GARCH_impr_%
AAPL	-0.754	9.511
EBAY	-0.478	8.500
MCD	0.516	0.960
MSFT	-0.131	7.846
NVDA	-0.423	7.579
SBUX	-0.400	9.498



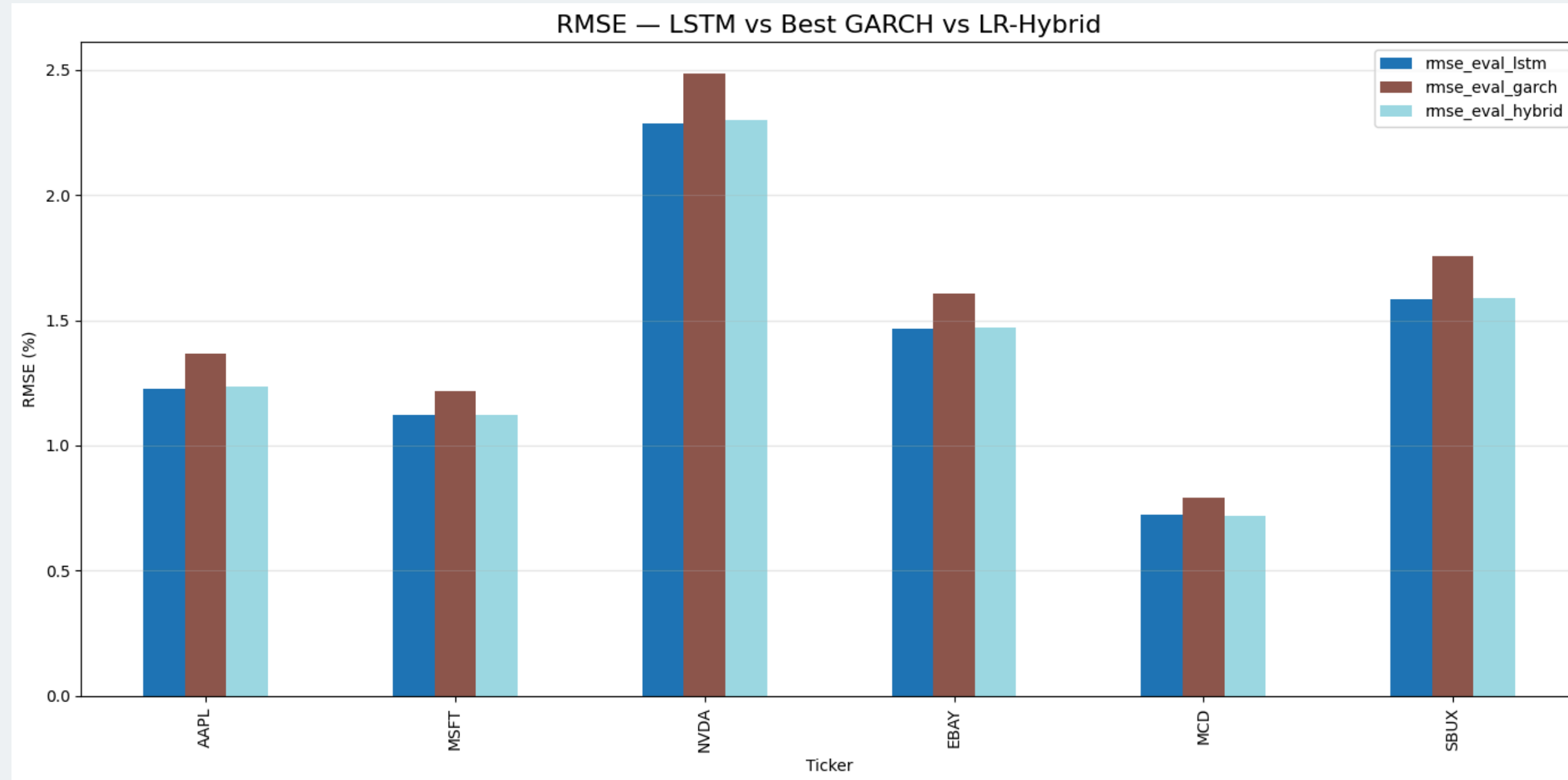
Results: Hybrid Models-Ridge regression

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- Compared to LSTM, the hybrid delivers very similar accuracy, usually within $\pm 0.5\%$, confirming stable performance across assets.
- Ridge regression tends to lean more heavily on GARCH weights, but still benefits from adding a small LSTM component.
- Overall, the ridge hybrid is robust and reliable, offering a strong middle ground between GARCH's stability and LSTM's flexibility.

AAPL:	Best GARCH=GARCH(1,2)	$\beta_{\text{lstm}}=0.032$	$\beta_{\text{garch}}=0.738$	eval RMSE	LSTM-1.2279	GARCH-1.3672	HYBRID-1.2372
MSFT:	Best GARCH=GARCH(2,2)	$\beta_{\text{lstm}}=0.373$	$\beta_{\text{garch}}=0.514$	eval RMSE	LSTM-1.1206	GARCH-1.2176	HYBRID-1.1221
NVDA:	Best GARCH=GARCH(1,1)	$\beta_{\text{lstm}}=0.020$	$\beta_{\text{garch}}=0.790$	eval RMSE	LSTM-2.2880	GARCH-2.4861	HYBRID-2.2977
EBAY:	Best GARCH=GARCH(1,2)	$\beta_{\text{lstm}}=0.223$	$\beta_{\text{garch}}=0.616$	eval RMSE	LSTM-1.4644	GARCH-1.6081	HYBRID-1.4714
MCD:	Best GARCH=GARCH(1,1)	$\beta_{\text{lstm}}=0.149$	$\beta_{\text{garch}}=0.614$	eval RMSE	LSTM-0.7252	GARCH-0.7924	HYBRID-0.7214
SBUX:	Best GARCH=GARCH(2,1)	$\beta_{\text{lstm}}=0.513$	$\beta_{\text{garch}}=0.399$	eval RMSE	LSTM-1.5842	GARCH-1.7574	HYBRID-1.5905

Linear regression

- The LR hybrid consistently improves over the best GARCH model, reducing RMSE by about 7–10% across all stocks.
- Compared to LSTM, the hybrid produces very similar accuracy, usually within $\pm 0.5\%$, confirming that linear blending is stable but rarely superior.
- The learned weights lean heavily toward GARCH ($\beta \approx 0.4\text{--}0.8$) rather than LSTM, meaning the hybrid mainly smooths GARCH using a small LSTM correction.
- Overall, the LR hybrid is simple, interpretable, and reliable, but does not meaningfully outperform LSTM on any stock.

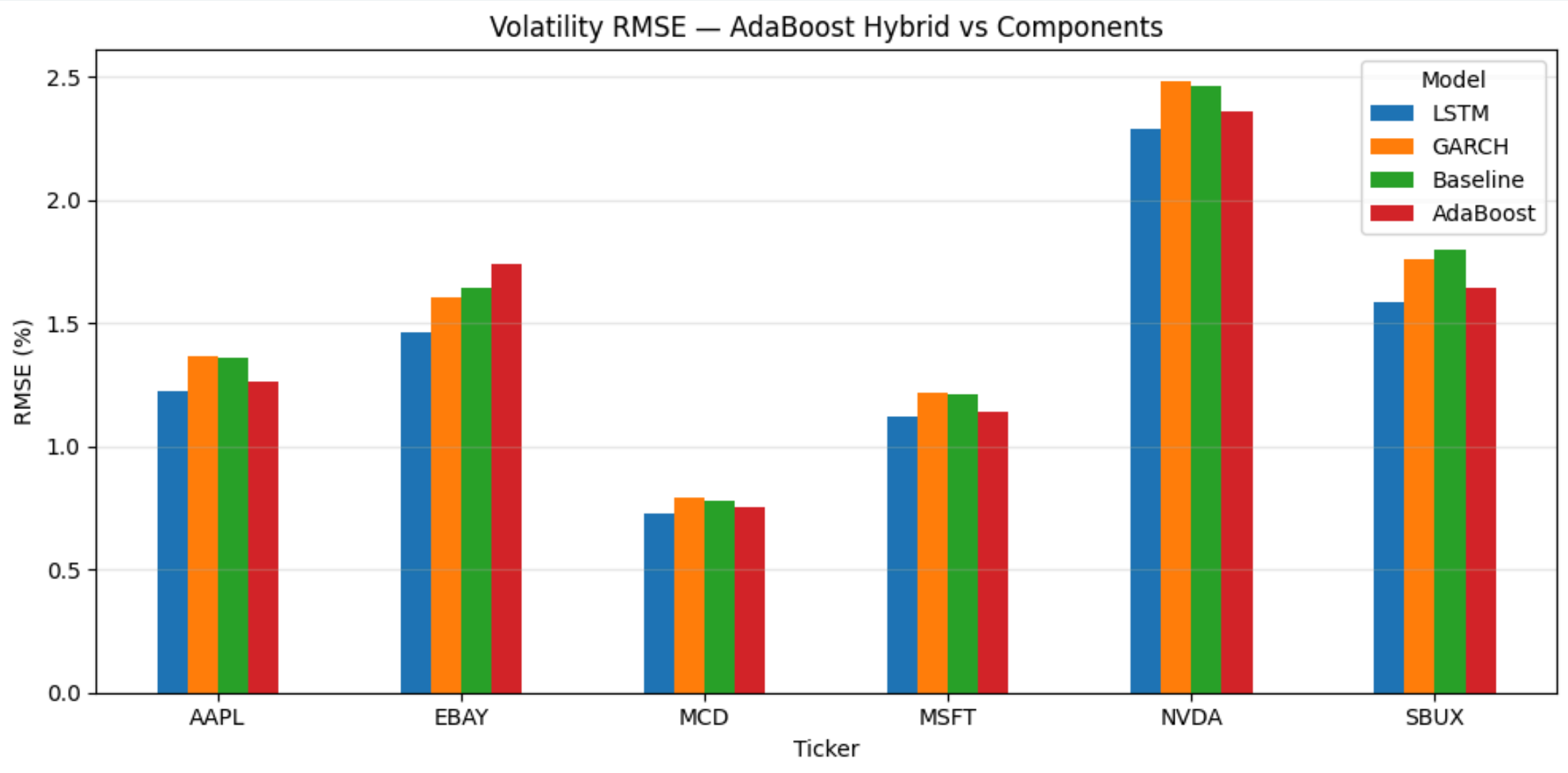


AAPL: OLS $\beta_{\text{lstm}}=0.032$, $\beta_{\text{garch}}=0.738$ | RMSE — LSTM 1.2279, GARCH 1.3672, HYBRID 1.2372
MSFT: OLS $\beta_{\text{lstm}}=0.373$, $\beta_{\text{garch}}=0.514$ | RMSE — LSTM 1.1206, GARCH 1.2176, HYBRID 1.1221
NVDA: OLS $\beta_{\text{lstm}}=0.020$, $\beta_{\text{garch}}=0.790$ | RMSE — LSTM 2.2880, GARCH 2.4861, HYBRID 2.2977
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MCD: OLS $\beta_{\text{lstm}}=0.149$, $\beta_{\text{garch}}=0.614$ | RMSE — LSTM 0.7252, GARCH 0.7924, HYBRID 0.7214
SBUX: OLS $\beta_{\text{lstm}}=0.513$, $\beta_{\text{garch}}=0.399$ | RMSE — LSTM 1.5842, GARCH 1.7574, HYBRID 1.5905

Adaboost

AdaBoost meta-learner (non-linear hybrid)

- AdaBoost improves over GARCH but is usually weaker than LSTM.
- RMSE typically sits between LSTM and GARCH, rarely beating LSTM.
- Model relies mostly on LSTM features (~80%), with little contribution from GARCH or Rolling Std.
- Non-linear but inconsistent, offering small gains on some stocks but less reliable than linear hybrids.



Ticker	LSTM	GARCH	Baseline	AdaBoost
AAPL	1.2279	1.367195	1.359773	1.26224
EBAY	1.46439	1.608082	1.647418	1.742203
MCD	0.725168	0.79242	0.776715	0.750794
MSFT	1.12059	1.217583	1.211106	1.140884
NVDA	2.288001	2.486096	2.465012	2.358849
SBUX	1.584168	1.757423	1.796908	1.642625

Feature importances
[LSTM, GARCH, RollingStd]: [0.81931866 0.11979231 0.06088902]

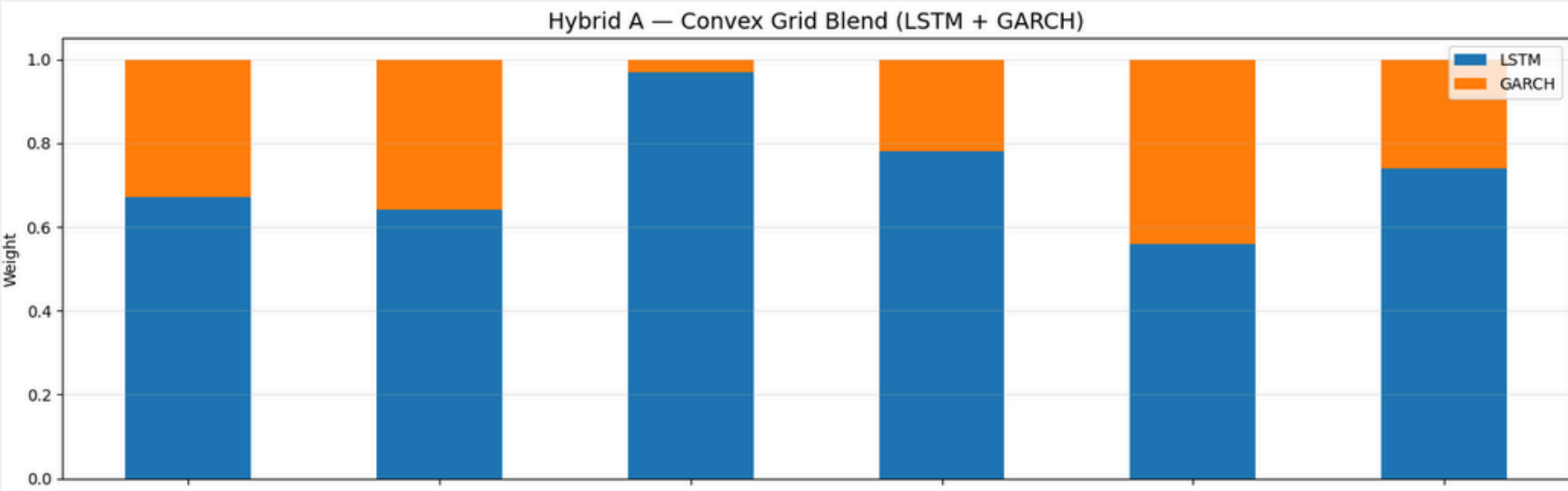
Adaboost

AdaBoost meta-learner (non-linear hybrid)

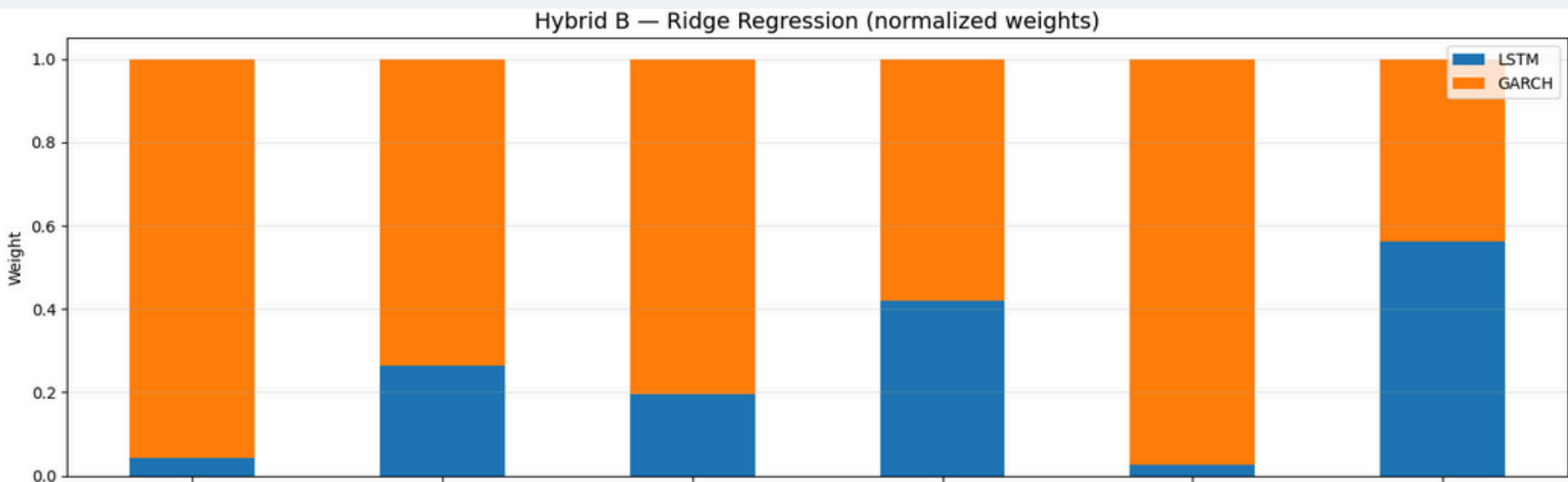
- AdaBoost improves over GARCH but is usually weaker than LSTM.
- RMSE typically sits between LSTM and GARCH, rarely beating LSTM.
- Model relies mostly on LSTM features (~80%), with little contribution from GARCH or Rolling Std.
- Non-linear but inconsistent, offering small gains on some stocks but less reliable than linear hybrids.

AAPL: Best GARCH=GARCH(1,2) | RMSE — LSTM 1.2279 | GARCH 1.3672 | Base 1.3598 | AdaBoost 1.2622
MSFT: Best GARCH=GARCH(2,2) | RMSE — LSTM 1.1206 | GARCH 1.2176 | Base 1.2111 | AdaBoost 1.1409
NVDA: Best GARCH=GARCH(1,1) | RMSE — LSTM 2.2880 | GARCH 2.4861 | Base 2.4650 | AdaBoost 2.3588
EBAY: Best GARCH=GARCH(1,2) | RMSE — LSTM 1.4644 | GARCH 1.6081 | Base 1.6474 | AdaBoost 1.7422
MCD: Best GARCH=GARCH(1,1) | RMSE — LSTM 0.7252 | GARCH 0.7924 | Base 0.7767 | AdaBoost 0.7508
SBUX: Best GARCH=GARCH(2,1) | RMSE — LSTM 1.5842 | GARCH 1.7574 | Base 1.7969 | AdaBoost 1.6426

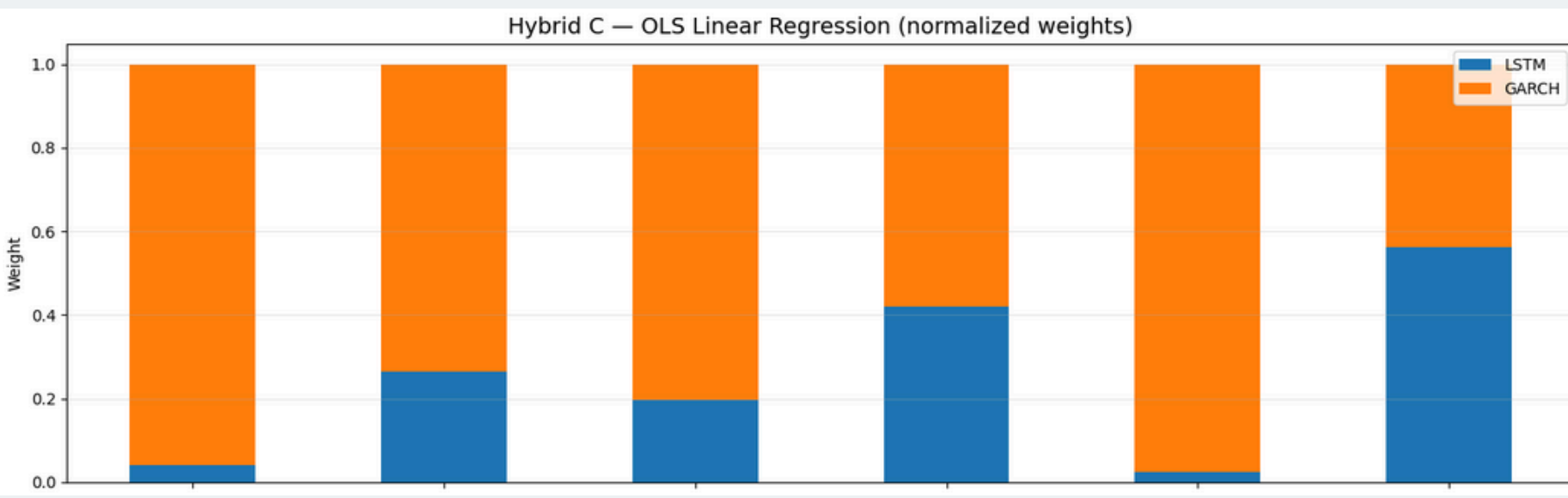
Convex grid blending gives most of the weight to LSTM, with GARCH acting as a smaller correction across all tickers.



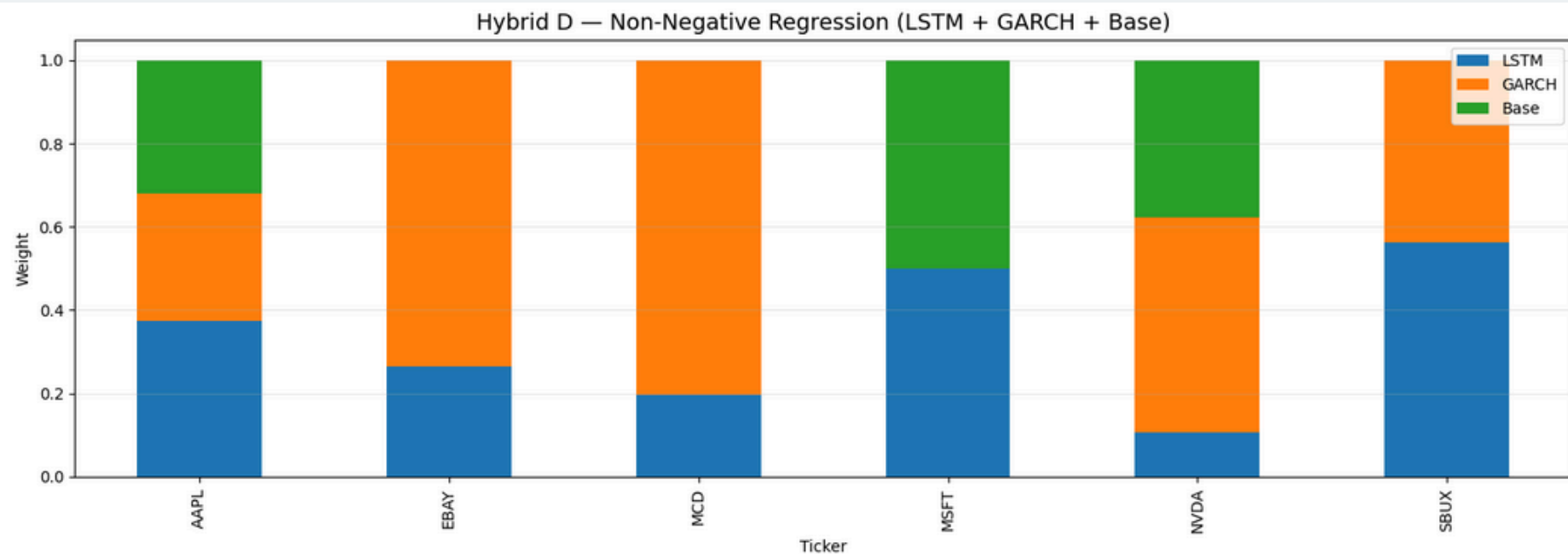
Ridge regression leans heavily on GARCH, while only partially loading on LSTM where it helps most.



Linear regression shows a similar pattern, again favoring GARCH with modest LSTM contributions for a few stocks.



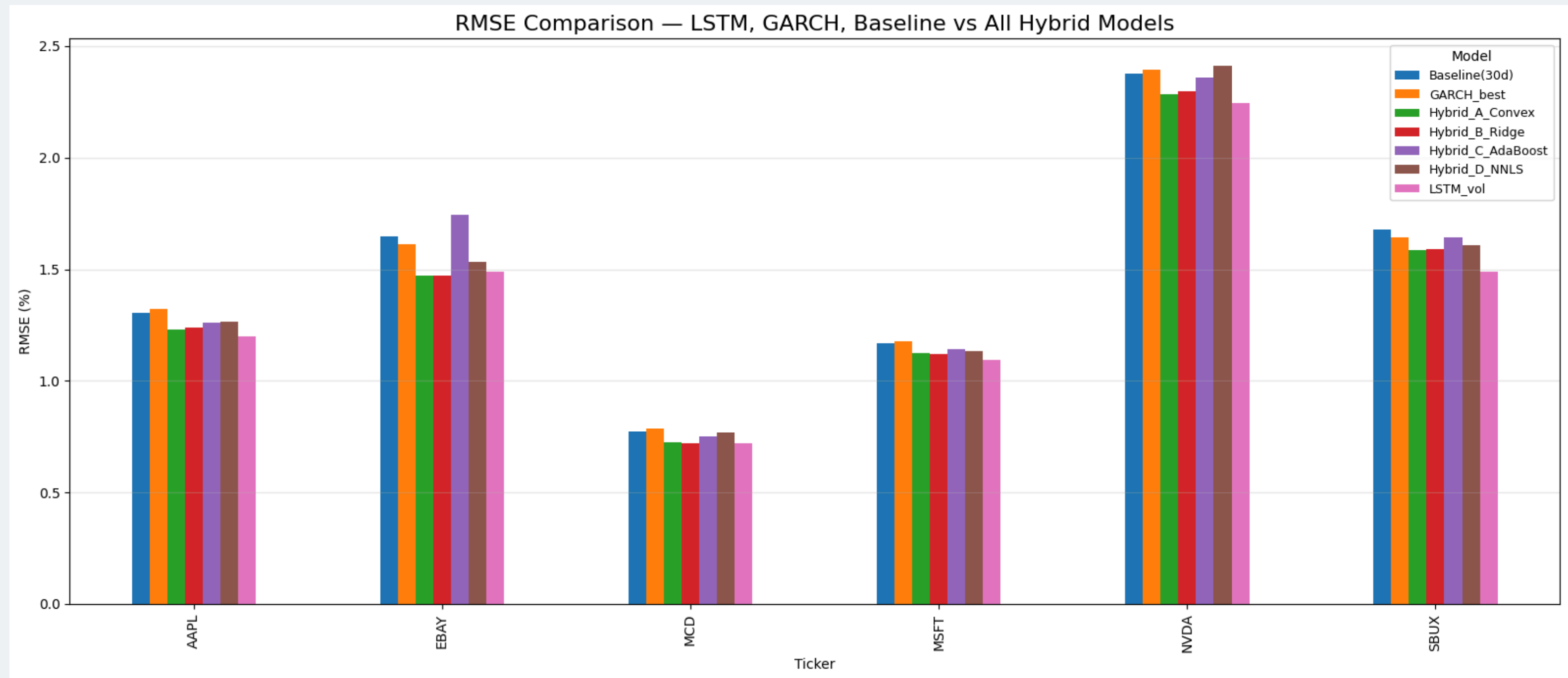
Non-negative regression mixes all three models, assigning complementary weights to LSTM, GARCH, and the baseline depending on the ticker.



Discussion: Model behavior and insights

The RMSE comparison shows that LSTM_vol consistently delivers the lowest error across all six stocks, improving on the 30-day baseline and best GARCH by roughly 5–11%.

Hybrid models (Convex, Ridge, AdaBoost, NNLS) typically lie between LSTM and the traditional methods, narrowing the gap but rarely outperforming the standalone LSTM, which captures most of the exploitable volatility structure.



Key findings:

- GARCH models provide a solid econometric baseline and capture volatility clustering.
- LSTM models learn nonlinear and regime-shift patterns, achieving the lowest RMSE on stocks but still underreacting to extreme volatility spikes.
- Hybrid models slightly outperform LSTM on EBAY and MCD, but do not dominate across all stocks.
- Combining GARCH and LSTM can improve accuracy for specific stocks and offers more stable, diversified forecasts.

Strengths:

- Compared multiple model families: ARCH, GARCH, LSTM, Hybrid 23-year dataset → robust evaluation
- Rolling one-step-ahead setup mirrors real market use
- Hybrid modeling adds practical and creative value

Limitations:

- Used only past returns; no macro features Simple LSTM architecture limits deep pattern learning
- Hybrid weights estimated on limited test data

Conclusion

The results are consistent across all six tickers: models that use LSTM forecasts clearly outperform pure GARCH and the simple 30-day baseline.

- LSTM_vol has the lowest RMSE for AAPL, MSFT, NVDA and SBUX, and is essentially tied with the best hybrid for MCD, only losing to Hybrid_B/Hybrid_A on EBAY.
- Hybrid_A (Convex LSTM+GARCH) always improves on pure GARCH, stays close to LSTM, and slightly beats it for EBAY, offering a good balance of structure and flexibility.
- GARCH_best and the 30-day baseline are reasonable econometric benchmarks but are consistently less accurate than LSTM-based models.
- Other hybrids (Ridge, NNLS) perform similarly to Hybrid_A with stock-specific differences, while AdaBoost is more erratic and sometimes worse than GARCH.

Overall conclusion: LSTM-driven models—alone or combined with GARCH—provide the most accurate and robust volatility forecasts in this experiment, with traditional baselines clearly lagging behind.

Thank You!

Reference

Applied econometric time series / Walter Enders Enders, Walter, 1948-
<https://id.oclc.org/worldcat/entity/E39PBJwdfqftQVgP6tbWTPxJjC>.

Dr. Sachin Kumar/Time series analysis CS362 / Generalised ARCH Lecture 24/25